

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461024.7519 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.061 +/- 0.042
Transit depth (Rp/Rs)^2	0.37 +/- 0.51 [%]
Orbital Inclination (inc)	82.07 +/- 2.9
Airmass coefficient 1 (a1)	1.038 +/- 0.084
Airmass coefficient 2 (a2)	-0.015 +/- 0.068
Scatter in the residuals of the lightcurve fit is	8.06 %
Best Comparison Star	#3 - [449, 332]
Optimal Aperture	2.13
Optimal Annulus	6.73
Transit Duration (day)	0.1138 +/- 0.0086

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.061 ± 0.042 vs 0.1178 (deviation 1.35σ)
Duration fit vs published	0.1138 d vs 0.125 d (deviation 1.3σ)
Mid-time O–C	2.0 min
Depth SNR	1.5
Residual scatter	8.06 %

Light curve

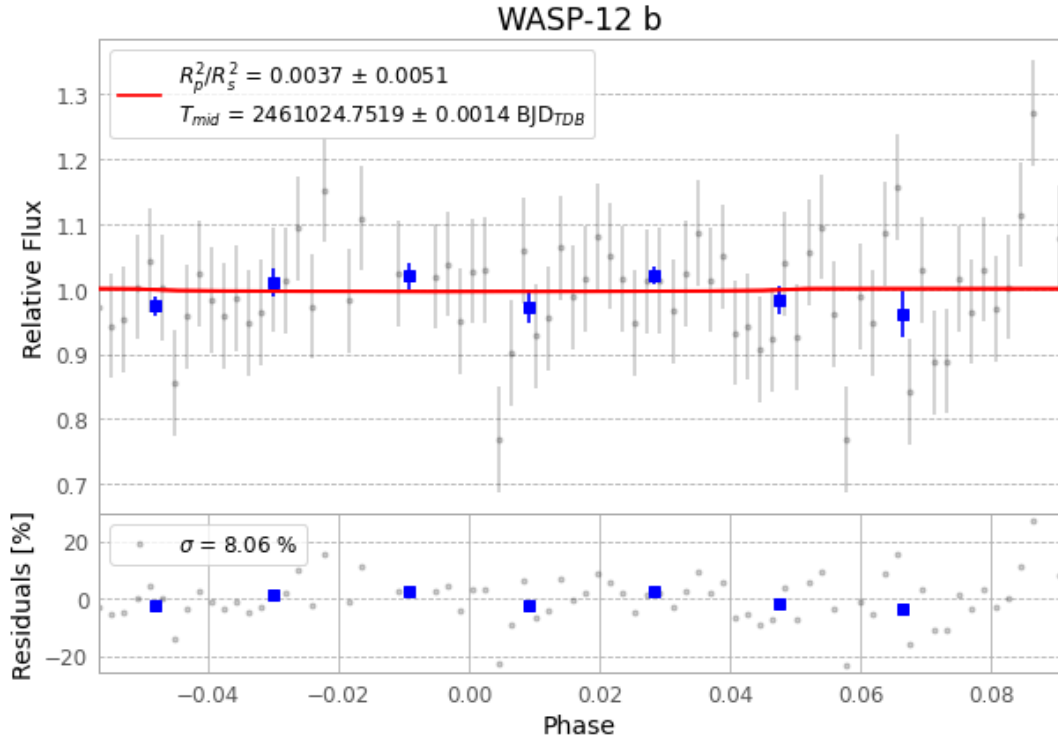


Figure 1 — FinalLightCurve_WASP-12 b_2025-12-14.png (SHA-256 2a184246e24e1ff5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [387, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 97374ffa7d764e3e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

81 FITS frames (53 MB), WASP-12251215042415.FITS → WASP-12251215082415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-251215.log (1d534c2b2e87...), FinalLightCurve_WASP-12 b_2025-12-14.png (2a184246e24e...), FinalLightCurve_WASP-12 b_2025-12-14.csv (39b57fdd9faf...)

Compute resources

Wall time 165.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:08Z.